

Book / Reference

Formula Reference

Every core formula in one place, straight from the game's balance config.

 Updated 2026-09-25

Progression

Level threshold (cumulative XP):

$$\text{xpForLevel}(n) = \begin{cases} \lfloor 300 \cdot n^{1.5} \rfloor & n \leq 75 \\ \lfloor \text{xp}(75) \cdot (n/75)^{1.8} \rfloor & 75 < n \leq 100 \\ \lfloor \text{xp}(100) \cdot (n/100)^{2.5} \rfloor & n > 100 \end{cases}$$

You are the highest level whose threshold you have crossed. Each level up pays **+5 gold**. The daily reward grants **10 XP**. The XP bar displays progress within the current level and resets on level up.

Energy

Maximum energy by level (regen is **+1 per minute**):

$$E_{\max}(\text{level}) = \min(300, 100 + 5(\text{level} - 1))$$

Work

Units produced per block of energy:

$$u_0 = \left\lfloor 2 \cdot \left(1 + 0.02 S_{\text{eco}} + 0.015 S_{\text{edu}} + 0.005(\text{level} - 1)\right) \cdot m_{\text{company}} \cdot \left(1 + \frac{b_{\text{region}}}{100}\right) \right\rfloor$$

$$u = \max(1, \lfloor u_0 \cdot m_{\text{industry}} \cdot m_{\text{crowd}} \rfloor)$$

$m_{\text{industry}} = 1 + 0.25 \cdot \frac{\text{industry}}{10} \cdot \frac{\text{electricity}}{10} \cdot \text{crown} + \text{ideology} + \text{convoy}$ is the region's Industry index (scaled by its electricity), plus a flat +0.15 under Socialism and up to +0.10 from a Mobile Workshop in your convoy; m_{crowd} is the region-capacity factor (below 1 when the region is crowded, above 1 for pioneers). A country whose capital fell in the last 30 days works at $\times 0.9$. Both floors drop fractions: a

bonus only shows once it pushes u_0 or u past the next whole number – a +10% resource bonus on $u_0 = 6.6$ still yields 6.

Here **crown** is 1.30, 1.20 or 1.10 in the three cities ranked highest by that day's City Score, and 1 everywhere else. Ideology, convoy and Premium are never multiplied by it.

Units in one shift (each block costs **10** energy):

$$\text{units} = \left\lfloor \frac{\text{energy} \cdot u}{10} \right\rfloor$$

Company multiplier m_{company} :

$$m_{\text{RAW}} = 1 + 0.01 (\text{level} - 1), \quad m_{\text{MFG}} = \text{tierMult} \cdot (1 + 0.02 (\text{level} - 1))$$

Work XP: **4 per 10-energy block**.

Training

$$\text{cost}(n) = n \text{ gold}, \quad \text{time}(n) = \text{round}\left(10 \cdot n \cdot 1.1^{n-1} \cdot \frac{1}{1+0.01 S_{\text{edu}}}\right) \text{ seconds}$$

Tax

$$\text{tax} = \lfloor \text{amount} \cdot \text{rate} \rfloor \quad (\text{minimum } 1 \text{ on any positive taxed amount})$$

Rates are clamped to **0–50%**.

War

Damage of one strike (all current energy is spent, minimum 10):

$$\text{damage} = \text{weapons} \cdot (5 + S_{\text{str}}) \cdot m_{\text{tier}}$$

Tier multiplier m_{tier} : **1.0 / 1.5 / 2.0 / 2.5 / 3.0** for T1–T5. Fight XP: **20 per 10 energy**. Wars last **24 h**; ties go to the defender.

Every hit is then rolled (Update 24) – a uniform swing of $\pm 10\%$, and a 5% chance of a critical at $\times 1.5$:

$$\text{dealt} = \max(1, \lfloor \text{damage} \cdot s \cdot c \rfloor), \quad s \sim U(0.9, 1.1), \quad c = 1.5 \text{ with } p = 0.05, \text{ else } 1$$

Flight

$minutes = round(10 + km \cdot 0.0235 \cdot f_{tier})$

Tier factor f_{tier} : 1.0 / 0.75 / 0.55 / 0.38 / 0.25 for T1-T5.

Ideology (12 September 2026)

A state ideology adds a flat share on one axis in every region of the state, independent of the indexes and of electricity:

$m = 1 \pm (\text{region term} + \text{ideology}), \quad \text{ideology} = \text{its share on its axis, else } 0$

The shares: Nationalism 0.10, Socialism 0.15, Liberalism 0.15, Conservatism 0.25.

Nationalism → damage, Socialism → industry, Liberalism → training time (subtracted), Conservatism → health regen.

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